General Relativity: A Detailed Derivation of Einstein's Field Equations

Khin Josmar V. Misuari*

Department of Physics, University of San Carlos, Cebu City, Philippines

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In 1915, Albert Einstein finalized his general theory of relativity, a framework that generalizes Newtonian gravity by describing gravity not as a force, but as the manifestation of curved spacetime. In this document, we present a detailed, step-by-step derivation of Einstein's field equations that shall serve as a supplement in following the physical-motivation approach of standard General Relativity textbooks such as Carroll's *Spacetime and Geometry*, as well as Misner, Thorne, and Wheeler's *Gravitation*. We begin with Poisson's equation for Newtonian gravity, then promote the mass density to the full energy-momentum tensor $T_{\mu\nu}$ and demand general covariance. We then invoke the contracted Bianchi identity to construct the unique symmetric, divergence-free geometric tensor and fix the proportionality constant by requiring the equations to reproduce Poisson's equation for slow-moving particles in a weak, static gravitational field. The advantage of this approach over the more formal Einstein-Hilbert variational formulation is that it builds directly on physical principles and mirrors Einstein's own historical reasoning, making it an ideal starting point for students new to general relativity. The reader is assumed to be familiar with the tensor formulation of special relativity, Newtonian gravitation, and the basics of pseudo-Riemannian geometry.

I. INTRODUCTION

General relativity (GR) is Einstein's theory of gravitation, in which gravity is not a force but rather the curvature of a four-dimensional spacetime manifold sourced by matter and energy [1–3]. Since its publication in 1915, GR has passed every experimental test devised,

* kj.misuari@gmail.com

from the precession of Mercury's perihelion [1] to the direct detection of gravitational waves by the LIGO–Virgo Collaboration [4]. General relativity forms one of the two foundational pillars of modern physics alongside quantum mechanics.

At the heart of GR are Einstein's field equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \tag{1}$$

where $G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$ is the Einstein tensor, Λ is the cosmological constant, and $T_{\mu\nu}$ is the energy-momentum tensor. These ten coupled, nonlinear partial differential equations encode, in the words of Wheeler [5]: *"Spacetime tells matter how to move; matter tells spacetime how to curve."*

There are two common ways to arrive at eq. (1). The more formal way proceeds via the Einstein-Hilbert action and a variational principle [2, 6]. The approach taken here follows the physical-motivation path: we build the field equations directly from Poisson's equation and the requirements of general covariance and energy-momentum conservation, mirroring Einstein's own historical reasoning. In this document, we will use the mostly positive metric signature $(-1, +1, +1, +1)$ and will not set $c = 1$.

This document is organized as follows. In section II we briefly discuss Poisson's equation and motivate a covariant generalization. In section III we use the Bianchi identity to construct the correct geometric tensor. In section IV we fix the proportionality constant by taking the Newtonian limit. We then summarize the discussion in section V.

II. POISSON'S EQUATION AND COVARIANT GENERALIZATION

In Newtonian mechanics, the gravitational potential ϕ is governed by Poisson's equation,

$$\nabla^2 \phi = 4\pi G \rho, \tag{2}$$

where ρ is the mass density and G is Newton's gravitational constant. Our goal is to find a relativistically covariant generalization of eq. (2).

In general, the rest-mass density ρ is not Lorentz invariant; under a boost it mixes with momentum density and pressure. To ensure the laws of physics remain the same for all observers, the momentum and pressure components that naturally arise during a Lorentz boost must also act as sources of gravity. The appropriate covariant generalization of the source is therefore the full energy-momentum tensor $T_{\mu\nu}$ [2, 3]. Similarly, the Newtonian potential ϕ must be generalized to an object encoding the full geometry of spacetime, the metric tensor $g_{\mu\nu}$; and the Laplacian ∇^2 must become a covariant differential operator:

$$\phi \longrightarrow g_{\mu\nu}, \qquad \rho \longrightarrow T_{\mu\nu}, \qquad \nabla^2 \longrightarrow \nabla^\sigma \nabla_\sigma. \tag{3}$$

A naive first guess is therefore:

$$\nabla^\sigma \nabla_\sigma \, g_{\mu\nu} = \kappa \, T_{\mu\nu}, \tag{4}$$

where κ is a constant to be determined. However, eq. (4) is immediately ill-posed: metric compatibility requires $\nabla_\sigma g_{\mu\nu} = 0$ so the left-hand side implies $T_{\mu\nu} = 0$, which is obviously not true in general. Thus, we must look elsewhere for the correct geometric tensor. To fix this problem, we must construct the LHS of the equation such that it satisfies the same property as the energy-momentum tensor on the RHS: symmetric ($T_{\mu\nu} = T_{\nu\mu}$) and conserved ($\nabla^\mu T_{\mu\nu} = 0$).

Conditions for the Newtonian Limit

Any valid field equation of gravity must reduce to eq. (2) in the appropriate limit. In other words, the "Newtonian limit" must be recovered by imposing three conditions [2]:

1. **Slow-moving particles** ($v \ll c$):

$$\frac{dx^i}{d\tau} \ll \frac{dx^0}{d\tau}, \tag{5}$$

which follows from $v \ll c \implies dx^i/dt \ll c \implies dx^i/d\tau \ll c\, dt/d\tau = dx^0/d\tau$.

2. **Weak gravitational field**: the metric is a small perturbation of flat spacetime,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \qquad |h_{\mu\nu}| \ll 1. \tag{6}$$

3. **Static field**: all metric components are time-independent,

$$\partial_0 \, g_{\mu\nu} = \partial_0 \, h_{\mu\nu} = 0. \tag{7}$$

III. THE EINSTEIN TENSOR VIA THE BIANCHI IDENTITY

Following the discussion in section II, we require a symmetric, divergence-free, rank-2 tensor built from the metric and its derivatives up to second order. One might then think of the full Riemann curvature tensor $R^{\rho}{}_{\mu\sigma\nu}$, but it carries too many indices. Its contraction however, the Ricci tensor $R_{\mu\nu}$, is a natural candidate since it is symmetric by construction and carries the correct index structure to match $T_{\mu\nu}$. Hence, an educated guess would be:

$$R_{\mu\nu} = \kappa\, T_{\mu\nu}, \tag{8}$$

However, this doesn't satisfy the conservation or divergence-free requirement because in general, $\nabla^{\mu} R_{\mu\nu} \neq 0$. In fact, the contracted Bianchi identity gives [2, 3]:

$$\nabla^{\mu} R_{\mu\nu} = \tfrac{1}{2}\nabla_{\nu} R. \tag{9}$$

However, from this same identity, we can construct a conserved tensor. Because $\nabla^{\mu} g_{\mu\nu} = 0$ by metric compatibility, $\frac{1}{2}\nabla_{\nu} R = \frac{1}{2}\nabla^{\mu}(g_{\mu\nu} R)$, so:

$$\begin{aligned} \nabla^{\mu} R_{\mu\nu} = \frac{1}{2}\nabla_{\nu} R \qquad &\Longrightarrow \qquad \nabla^{\mu} R_{\mu\nu} = \frac{1}{2}\nabla^{\mu} g_{\mu\nu} R \\ &\Longrightarrow \qquad \nabla^{\mu}\left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}\right) = 0 \end{aligned}$$

The combination $R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu}$ is the *Einstein tensor* $G_{\mu\nu}$. Additionally, metric compatibility further permits adding a term proportional to $g_{\mu\nu}$ while still maintaining the conservation:

$$\nabla^{\mu}\big(R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} + \Lambda\, g_{\mu\nu}\big) = 0, \tag{10}$$

for any constant Λ. Hence we have our primary candidate. The tensor $R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu} + \Lambda g_{\mu\nu}$ is symmetric, divergence-free as shown in eq. (10), and built from at most second derivatives of the metric, exactly the properties required to match $T_{\mu\nu}$ on the right-hand side. Our proposed field equation is thus:

$$\boxed{R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} + \Lambda\, g_{\mu\nu} = \kappa\, T_{\mu\nu}.} \tag{11}$$

IV. FIXING THE CONSTANT: THE NEWTONIAN LIMIT

If eq. (11) is our sought generalized gravitational field equation, it must reduce to Newtonian gravity under the Newtonian-limit conditions of section II. In other words, we now must fix κ by demanding that eq. (11) reduces to Poisson's equation (eq. (2)) in the appropriate limit.

Alternative Form

Before we take the Newtonian limit, let us first simplify eq. (11) by taking its trace. Multiplying by $g^{\mu\nu}$ yields:

$$g^{\mu\nu}R_{\mu\nu} - \frac{1}{2}Rg^{\mu\nu}g_{\mu\nu} + \Lambda g^{\mu\nu}g_{\mu\nu} = \kappa g^{\mu\nu}T_{\mu\nu} \quad \Longrightarrow \quad R - \frac{4}{2}R + 4\Lambda = \kappa T$$
$$\Longrightarrow \quad R = -\kappa T + 4\Lambda$$

Substituting this into eq. (11) gives the equivalent form:

$$R_{\mu\nu} - \frac{1}{2}(-\kappa T + 4\Lambda)g_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$$
$$R_{\mu\nu} - \Lambda g_{\mu\nu} = \kappa\left(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu}\right) \tag{12}$$

Energy-Momentum Tensor of Dust

Let us consider a slow-moving ($v \ll c$), massive particle or dust (like a planet or star). In its rest frame, its four-velocity is $U^\mu = (c, \mathbf{0})$. The energy-momentum tensor for a dust is [2]:

$$T_{\mu\nu} = \rho U_\mu U_\nu \tag{13}$$

The normalization condition $g^{\mu\nu}U_\mu U_\nu = -c^2$ reduces, to leading order in $h_{\mu\nu}$, to $\eta^{\mu\nu}U_\mu U_\nu = -c^2$, giving $U_0U_0 = c^2$. Hence:

$$T_{00} = c^2\rho \tag{14}$$

To leading order in $h_{\mu\nu}$, $T = g^{\mu\nu}T_{\mu\nu} \approx \eta^{\mu\nu}T_{\mu\nu}$. Hence:

$$T = \eta^{00}T_{00} + \eta^{ij}T_{ij}$$
$$T \approx -c^2\rho \tag{15}$$

The $(\mu, \nu) = (0, 0)$ Component

Taking the $(0, 0)$ component of eq. (12) and using eq. (14) and eq. (15), we get:

$$\begin{aligned} R_{00} - \Lambda\eta_{00} &= \kappa\big(T_{00} - \tfrac{1}{2}T\,\eta_{00}\big) \\ &= \kappa\big(c^2\rho - \tfrac{1}{2}(-c^2\rho)(-1)\big)\,, \\ R_{00} - \Lambda\eta_{00} &= \tfrac{1}{2}\kappa c^2\rho. \end{aligned} \tag{16}$$

The Ricci component R_{00} expands as:

$$R_{00} = \partial_\sigma \Gamma^\sigma_{00} - \partial_0 \Gamma^\sigma_{\sigma 0} + \Gamma^\sigma_{\sigma\lambda}\Gamma^\lambda_{00} - \Gamma^\sigma_{0\lambda}\Gamma^\lambda_{0\sigma}. \tag{17}$$

The second term vanishes by the static condition (eq. (7)); the last two terms are second order in $h_{\mu\nu}$ and are neglected. Hence:

$$R_{00} \approx \partial_\sigma \Gamma^\sigma_{00} = \partial_0 \Gamma^0_{00} + \partial_i \Gamma^i_{00} = \partial_i \Gamma^i_{00} \tag{18}$$

The relevant Christoffel symbol, linearized in $h_{\mu\nu}$ and using eq. (7), evaluates to:

$$\Gamma^i_{00} = \tfrac{1}{2} g^{ij}(\partial_0 g_{j0} + \partial_0 g_{0j} - \partial_j g_{00}) \approx -\tfrac{1}{2}\eta^{ij}\partial_j h_{00} = -\tfrac{1}{2}\partial^i h_{00}. \tag{19}$$

Substituting eq. (19) into eq. (18):

$$\begin{aligned} R_{00} &\approx \partial_i \Gamma^i_{00} = \partial_i \left(-\frac{1}{2}\partial^i h_{00}\right) \\ &\approx -\frac{1}{2}\partial_i \partial^i h_{00} \\ R_{00} &\approx -\frac{1}{2}\nabla^2 h_{00} \end{aligned} \tag{20}$$

Substituting eq. (20) into eq. (16), we get:

$$\begin{aligned} R_{00} - \Lambda\eta_{00} = \frac{1}{2}\kappa c^2 \rho \quad &\Longrightarrow \quad -\frac{1}{2}\nabla^2 h_{00} - \Lambda\eta_{00} = \frac{1}{2}\kappa c^2\rho \\ &\Longrightarrow \quad \nabla^2 h_{00} = -\kappa c^2 \rho + 2\Lambda \end{aligned} \tag{21}$$

This *almost* looks like Poisson's equation, except we have to relate h_{00} to the gravitational potential ϕ. How do we do that?

Relating h_{00} to the Newtonian Potential

To connect h_{00} to ϕ, recall that the covariant form of the familiar Newton's second law for a freely-falling object, $m\frac{d^2x^i}{dt^2} = -m(\partial^i \phi)$, is the geodesic equation:

$$\frac{d^2x^\mu}{d\tau^2} = -\Gamma^\mu_{\rho\sigma}\frac{dx^\rho}{d\tau}\frac{dx^\sigma}{d\tau} = -\Gamma^\mu_{00}\frac{dx^0}{d\tau}\frac{dx^0}{d\tau} - 2\Gamma^\mu_{0j}\frac{dx^0}{d\tau}\frac{dx^j}{d\tau} - \Gamma^\mu_{ij}\frac{dx^i}{d\tau}\frac{dx^j}{d\tau} \tag{22}$$

The first condition of the Newtonian limit tells us that $\frac{dx^i}{d\tau} \ll \frac{dx^0}{d\tau}$, hence under the slow-motion condition (eq. (5)), the geodesic equation reduces to:

$$\frac{d^2x^\mu}{d\tau^2} \approx -\Gamma^\mu_{00}\left(\frac{dx^0}{d\tau}\right)^2. \tag{23}$$

Its spatial components give:

$$\frac{d^2x^i}{d\tau^2} = -\Gamma^i_{00}\frac{dx^0}{d\tau}\frac{dx^0}{d\tau} = \Gamma^i_{00}\frac{d(ct)}{d\tau}\frac{d(ct)}{d\tau} \quad \Longrightarrow \quad \frac{d^2x^i}{d\tau^2}\frac{d\tau^2}{dt^2} = -c^2\Gamma^i_{00} \tag{24}$$

Hence:

$$\frac{d^2x^i}{dt^2} = -c^2\left(-\frac{1}{2}\partial^i h_{00}\right) = -\partial^i\left(-\frac{1}{2}c^2 h_{00}\right) \tag{25}$$

Comparing with Newton's second law $\frac{d^2x^i}{dt^2} = -\partial^i\phi$, we must have:

$$\phi = -\frac{c^2}{2}h_{00} \quad \Longrightarrow \quad h_{00} = -\frac{2\phi}{c^2}. \tag{26}$$

Recovering Poisson's Equation

Substituting eq. (26) into eq. (21), we get:

$$\begin{aligned}\frac{\nabla^2\phi}{c^2} &= \tfrac{1}{2}\kappa c^2\rho - \Lambda \\ \nabla^2\phi &= \tfrac{1}{2}\kappa c^4\rho - \Lambda c^2.\end{aligned} \tag{27}$$

Comparing eq. (27) with Poisson's equation (eq. (2)), our proposed field equation reduces to the appropriate Newtonian limit provided that two things are true. First, the constant Λ must be sufficiently small enough such that it negligibly small in Newtonian mechanics, $\nabla^2\phi \approx \frac{1}{2}\kappa c^4\rho$. Second, the constant κ must be:

$$\tfrac{1}{2}\kappa c^4 = 4\pi G \quad \Longrightarrow \quad \boxed{\kappa = \frac{8\pi G}{c^4}.} \tag{28}$$

Substituting eq. (28) into eq. (11) yields **Einstein's field equations**:

$$\boxed{R_{\mu\nu} - \tfrac{1}{2}R\,g_{\mu\nu} + \Lambda\,g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}.} \tag{29}$$

This is a system of coupled, nonlinear partial differential equations relating the geometry of spacetime to the distribution of matter and energy. Because it's a rank 2 tensor, there are 16 components in total (4x4 matrix) and thus sixteen equations. However, because the field equation is symmetric, the twelve off-diagonal terms of the 4x4 matrix reduce to 6 non-trivial ones. Hence, there are only 10 non-trivial equations.

The first two terms on the left-hand side of the equation ($G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}R\,g_{\mu\nu}$) describe the geometry/curvature of spacetime; the right-hand side encodes the matter-energy content.

The coupling constant $8\pi G/c^4 \sim 10^{-43}\,\mathrm{N}^{-1}$ is extraordinarily small, reflecting the fact that enormous concentrations of mass-energy are required to produce observable curvature, which is why only astrophysical and cosmological objects exhibit non-negligible gravitational effects.

What about the third term $\Lambda\, g_{\mu\nu}$? One way of understanding its physical significance is by rewriting eq. (29) as:

$$\boxed{R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu} - \Lambda\, g_{\mu\nu}.} \tag{30}$$

Notice that in the absence of matter (in vacuum) where $T_{\mu\nu} = 0$, the equation reduces to $R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu} = -\Lambda\, g_{\mu\nu}$, implying that even in the absence of matter, there is a non-zero "vacuum" energy density that causes gravitational effects. Nowadays, we associate this constant Λ with the cosmological constant that describes the accelerated expansion of the universe. For a positive cosmological constant ($\Lambda > 0$), this behaves exactly like a negative pressure (dark energy) driving an accelerated de Sitter expansion, which aligns flawlessly with modern standard ΛCDM cosmology.

V. CONCLUSION

We have derived Einstein's field equations (eq. (29)) from first principles, beginning with Poisson's equation and imposing the requirements of general covariance and energy-momentum conservation. The key steps were: (i) recognizing that the relativistic source of gravity must be the full energy-momentum tensor $T_{\mu\nu}$; (ii) using the contracted Bianchi identity to identify the unique symmetric, divergence-free geometric tensor; and (iii) fixing the value of magnitude of the constant Λ and coupling constant $\kappa = 8\pi G/c^4$ by demanding recovery of Newtonian gravity in the weak-field, slow-motion limit.

The cosmological constant Λ, which emerges here as a free parameter consistent with the Bianchi identity, is interpreted in modern cosmology as the energy density of the quantum vacuum and drives the observed accelerated expansion of the universe [7].

From here, the natural next steps are the exact solutions of eq. (29): the Schwarzschild solution for static spherical symmetry [8], the Kerr solution for rotating black holes, linearized gravity and gravitational waves [4], and the Friedmann-Lemaître-Robertson-Walker cosmological solutions [2, 3].

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